

Calculus I Review

MATH 146 – Calculus II

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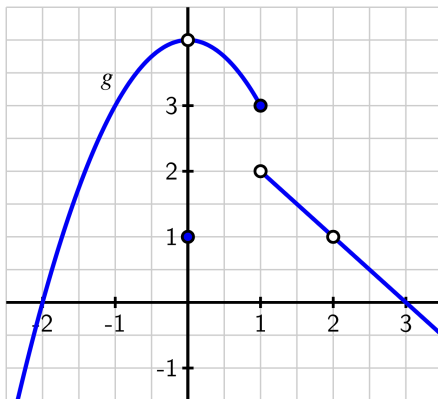
Limit

Given a function f , a fixed input $x = a$, and a real number L , we say that f **has limit** L as x **approaches** a , and write

$$\lim_{x \rightarrow a} f(x) = L$$

provided that we can make $f(x)$ as close to L as we like by taking x sufficiently close (but not equal) to a . If we cannot make $f(x)$ as close to a single value as we would like as x approaches a , then we say that f **does not have a limit** as x **approaches** a .

Limits Example



Left-Hand Limits

Left-Hand Limit

We say that f **has limit** L_1 **as** x **approached** a **from the left** and write

$$\lim_{x \rightarrow a^-} f(x) = L_1$$

provided that we can make the value of $f(x)$ as close to L_1 as we like by taking x sufficiently close to a , while always having $x < a$. We call L_1 the **left-hand limit of f as x approaches a** .

Right-Hand Limits

Right-Hand Limit

We say that f has limit L_2 as x approached a from the right and write

$$\lim_{x \rightarrow a^+} f(x) = L_2$$

provided that we can make the value of $f(x)$ as close to L_2 as we like by taking x sufficiently close to a , while always having $x > a$. We call L_1 the **right-hand limit of f as x approaches a** .

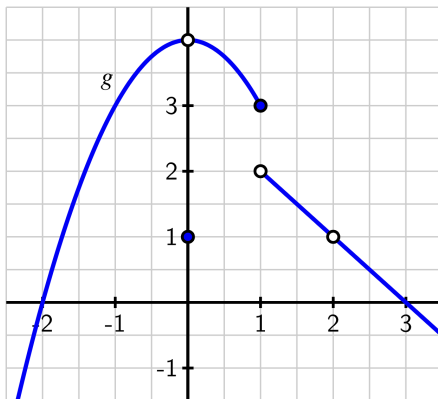
Left-Hand and Right-Hand Limits

Limit

A function f has limit L as $x \rightarrow a$ if and only if

$$\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

Limits Example



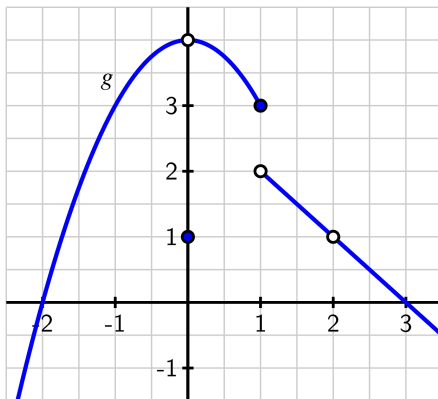
What does it mean for a function to be continuous?

Continuous at a Point

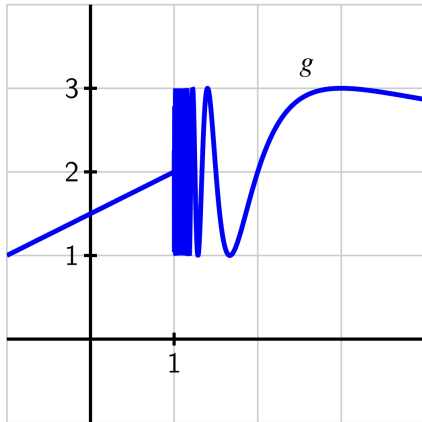
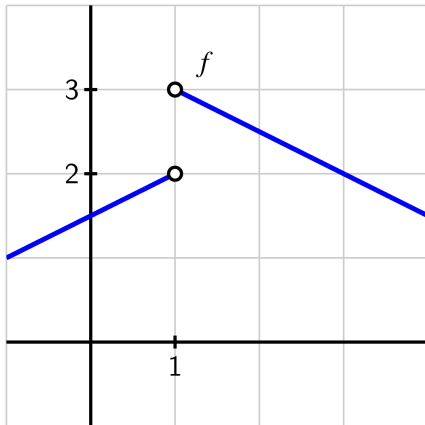
A function f is **continuous at** $x = a$ provided that

- f has a limit as $x \rightarrow a$,
- f is defined at $x = a$, and
- $\lim_{x \rightarrow a} f(x) = f(a)$.

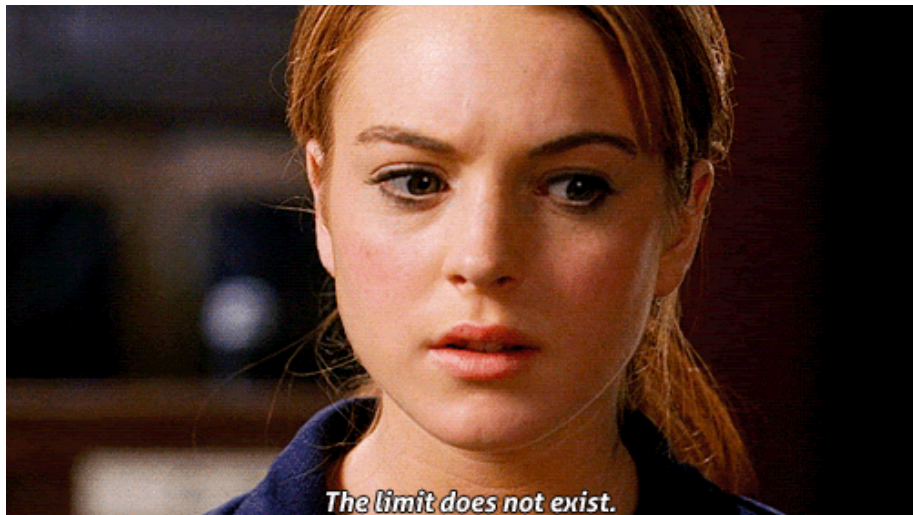
Limits Example



The Limit Does Not Exist!



The Limit Does Not Exist!



The Derivative Function

This leads us to the following definition:

The Derivative Function

Let f be a function and x a value in the function's domain. We define the **derivative of f** , a new function called f' , by the formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

$f'(x)$ is sometimes written as $\frac{d}{dx} f(x)$.

Application of the Derivative

Increasing Function

We say that f is **increasing on** (a, b) provided that the function is always rising as we move from left to right. That is, for any x and y in (a, b) , if $x < y$, then $f(x) < f(y)$.

Decreasing Function

We say that f is **decreasing on** (a, b) provided that the function is always falling as we move from left to right. That is, for any x and y in (a, b) , if $x < y$, then $f(x) > f(y)$.

Application of the Derivative

Increasing Function

Let f be a function that is differentiable on an interval (a, b) . If $f'(x) > 0$ for every x such that $a < x < b$, then f is **increasing on** (a, b) .

Decreasing Function

Let f be a function that is differentiable on an interval (a, b) . If $f'(x) < 0$ for every x such that $a < x < b$, then f is **decreasing on** (a, b) .

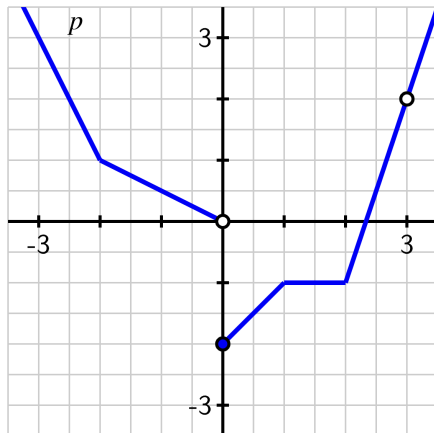
Differentiability

Differentiability

We say that a function f is **differentiable at** $x = a$ if it has a derivative at $x = a$.

- If f is differentiable at $x = a$, the f is continuous at $x = a$.
Equivalently, if f fails to be continuous at $x = a$, the f will not be differentiable at $x = a$.
- A function can be continuous at a point, but not be differentiable there. In particular, a function f is not differentiable at $x = a$ if the graph has a sharp corner (or *cusp*) at the point $(a, f(a))$.
- If f is differentiable at $x = a$, the f is *locally linear* at $x = a$. That is, when a function is differentiable, it looks linear when viewed up close because it resembles its tangent line there.

Example: Continuity and Differentiability



The Constant Multiple Rule

The Constant Multiple Rule

For any real number k , if $f(x)$ is a differentiable function with derivative $f'(x)$, then $\frac{d}{dx} [kf(x)] = kf'(x)$.

The Sum Rule

The Sum Rule

If $f(x)$ and $g(x)$ are differentiable functions with derivatives $f'(x)$ and $g'(x)$ respectively, then $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

The Power Rule

The Power Rule

For any nonzero real number n , if $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

The Product Rule

The Product Rule

If f and g are differentiable functions, then their product $P(x) = (f \cdot g)(x) = f(x) \cdot g(x)$ is also a differentiable function, and

$$P'(x) = f(x)g'(x) + g(x)f'(x).$$

The Quotient Rule

The Quotient Rule

If f and g are differentiable functions, then their quotient $Q(x) = \frac{f(x)}{g(x)}$ is also a differentiable function for all x where $g(x) \neq 0$, where

$$Q'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}.$$

The Chain Rule

The Chain Rule

If f and g are differentiable functions, then:

$$\frac{d}{dx}(f \circ g)(x) = \frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

Derivatives of Trigonometric Functions

- $\frac{d}{dx} \sin(x) = \cos(x).$
- $\frac{d}{dx} \cos(x) = -\sin(x).$
- $\frac{d}{dx} \tan(x) = \sec^2(x).$
- $\frac{d}{dx} \sec(x) = \sec(x) \tan(x).$
- $\frac{d}{dx} \csc(x) = -\csc(x) \cot(x).$
- $\frac{d}{dx} \cot(x) = -\csc^2(x).$

Derivatives of Inverse Trigonometric Functions

- $\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}, x \neq 0.$
- $\frac{d}{dx} \cos^{-1}(x) = -\frac{1}{\sqrt{1-x^2}}, x \neq 0.$
- $\frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2}.$
- $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{|x|\sqrt{x^2-1}}, x \neq 0, \pm 1.$
- $\frac{d}{dx} \csc^{-1}(x) = -\frac{1}{|x|\sqrt{x^2-1}}, x \neq 0, \pm 1.$
- $\frac{d}{dx} \cot^{-1}(x) = -\frac{1}{1+x^2}.$

Derivatives of Exponential & Logarithmic Functions

- $\frac{d}{dx}e^x = e^x.$
- $\frac{d}{dx}a^x = a^x \ln(a).$
- $\frac{d}{dx} \ln(x) = \frac{1}{x}, \quad x > 0.$
- $\frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)}, \quad x > 0.$

Antiderivatives

Antiderivatives

If $f(x)$ is a function, then an **antiderivative** of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$.

Integral Notation

Antiderivatives

If $f(x)$ is a function, then the most general antiderivative of $f(x)$ is denoted by:

$$\int f(x) dx$$

Example:

$$\int 2x dx = x^2 + C$$

Rules for Antiderivatives

Rules for Finding Antiderivatives:

- (Inverse) Power Rule:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \text{ for all } n \neq -1.$$

- Sum Rule:

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx.$$

- Constant Multiple Rule:

$$\int cf(x) dx = c \int f(x) dx.$$

Differential Equations

Differential Equations

A **differential equation** is an equation that relates an unknown function $y(x)$ and one or more of its derivatives.

A differential equation with an initial condition $y(x_0) = y_0$ is an **initial value problem**.